Assessments for the year

Organisation. Each exam has two specimen papers for practice. They indicate the expected methods; they can be given at the start of the sequence, even if not all the notions have been taught yet. The table shows the planned time of distribution. Versions A and B are reserved for assessment.

Due date	Scheduled for	Specimen papers
DS1	S9 · week of 21 September	From S2
DS2	S18 · week of 12 October	From S14
Common test	S28 · week of 16 November; time slot to be coordinated	From S14
DS3	S34 · week of 30 November	From S21
Mock 1	S45 · 14-16 January, 4-hour slot to be scheduled	From S36
DS4	S57 · week of 8 February	From S44
DS5	S68 · week of 22 March	From S52
Mock 2	S71 · 1-3 April, 4-hour slot to be scheduled	From S59
Final mock exam	S90 · week of 31 May, 4-hour slot to be scheduled	From S75

DS1 · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 1$ and, for every natural number n, $u_{n+1} = \frac{1}{2}u_n + 4$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Prove by induction that, for every $n \in \mathbb{N}$, $u_n = 8 - 7\left(\frac{1}{2}\right)^n$. (2 pt)

3. Deduce that, for every natural number n, $1 \leq u_n < 8$. (1 pt)

4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. A student claims: "There exists an integer n such that $u_n = 8$." Is this statement true? Justify. (0.5 pt)

Exercise 2 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{72 + v_n}$.

1. Calculate v_1 and v_2. Give exact values. (0.5 pt)

2. Prove by induction that all terms are defined and that, for every $n \in \mathbb{N}$, $0 \leq v_n \leq 9$. (1.5 pt)

3. Prove by induction that, for every $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Prove that, for every $n \in \mathbb{N}$, $v_n < 9$. (1 pt)

5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to state that $P(0)$ is true? Justify. (0.5 pt)

DS1 · Specimen 1 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+5\vec{\jmath}+9\vec{k}, \quad \vec{z}=4\vec{\imath}+5\vec{\jmath}+10\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (1.5 pt)

4. Express the vector $5\vec{u}-4\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped ABCDEFGH non-degenerate. The vertices E, F, G and H are the images respectively of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Set $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{11}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{11}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{11}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Prove that the lines (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. (1 pt)

4. Let O the centre of the parallelepiped. Prove that O lies on the line (AG). (1 pt)

5. Prove that the points I, J and K are not collinear. (1 pt)

DS1 · Specimen 2

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every natural number n, $u_{n+1} = \frac{1}{5}u_n + \frac{36}{5}$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Prove by induction that, for every $n \in \mathbb{N}$, $u_n = 9 - 7\left(\frac{1}{5}\right)^n$. (2 pt)

3. Deduce that, for every natural number n, $2 \leq u_n < 9$. (1 pt)

4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. A student claims: "There exists an integer n such that $u_n = 9$." Is this statement true? Justify. (0.5 pt)

Exercise 2 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{90 + v_n}$.

1. Calculate v_1 and v_2. Give exact values. (0.5 pt)

2. Prove by induction that all terms are defined and that, for every $n \in \mathbb{N}$, $0 \leq v_n \leq 10$. (1.5 pt)

3. Prove by induction that, for every $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (1.5 pt)

4. Prove that, for every $n \in \mathbb{N}$, $v_n < 10$. (1 pt)

5. Consider a property $P(n)$, defined for $n \in \mathbb{N}$. We have proved $P(1)$ and, for all $n \geq 1$, the implication $P(n) \Rightarrow P(n+1)$. Do these two results allow us to state that $P(0)$ is true? Justify. (0.5 pt)

DS1 · Specimen 2 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+7\vec{\jmath}+10\vec{k}, \quad \vec{z}=3\vec{\imath}+7\vec{\jmath}+11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (1.5 pt)

4. Express the vector $7\vec{u}-3\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped ABCDEFGH non-degenerate. The vertices E, F, G and H are the images respectively of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Set $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Prove that the lines (IJ) and (BD) are parallel. (1 pt)

3. Express $\overrightarrow{CG}$ in the basis, then determine whether $\overrightarrow{AB}$ and $\overrightarrow{CG}$ are collinear. Justify. (1 pt)

4. Let O the centre of the parallelepiped. Prove that O lies on the line (AG). (1 pt)

5. Prove that the points I, J and K are not collinear. (1 pt)

DS2 · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{3n^2+4n-1}{4n^2+3}$ & $v_n = \sqrt{n^2+4n} - n$.

1. Determine the limit of (u_n) showing how its expression is transformed. (1 pt)

2. Show that, for every integer $n \geq 1$, $v_n = \frac{4}{\sqrt{1+\frac{4}{n}}+1}$. (1 pt)

3. Deduce the limit of (v_n). (1 pt)

4. Define $w_n = 3n^2 - 4n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this statement true? Justify. (1 pt)

Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unjustified answer earns no marks. Limits of sequences are taken as n tends to $+\infty$.

1. If $u_n \to 2$ and $v_n \to -5$, then $\frac{u_n}{v_n} \to -\frac{2}{5}$. (1 pt)

2. The sequence $u_n = 2 + (-1)^n$ converges to 2. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. The sequence $w_n = \frac{\sqrt{n^2+3n}}{n}$ for $n \geq 1$ converges to 1. (1 pt)

5. For $u_n = \frac{2n+5}{n}$, we have $|u_n - 2| < 0.01$ as soon as $n > 500$. (1 pt)

DS2 · Specimen 1 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=2\vec{\imath}+6\vec{\jmath}+8\vec{k}, \quad \vec{z}=2\vec{\imath}+6\vec{\jmath}+9\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (1.5 pt)

4. Express the vector $\vec{u}-2\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped ABCDEFGH non-degenerate. The vertices E, F, G and H are the images respectively of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Set $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{18}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{18}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{18}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Prove that the lines (IJ) and (BD) are parallel. (1 pt)

3. Study the relative position of the lines (AB) and (CG) : are they parallel, intersecting or not coplanar? Justify. (1 pt)

4. Let O the parallelepiped's centre. Does point O lie in the plane (ABG) ? Justify. (1 pt)

5. Prove that the points I, J and K are not collinear. (1 pt)

DS2 · Specimen 2

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{2n^2+6n-1}{3n^2+1}$ & $v_n = \sqrt{n^2+6n} - n$.

1. Determine the limit of (u_n) showing how its expression is transformed. (1 pt)

2. Show that, for every integer $n \geq 1$, $v_n = \frac{6}{\sqrt{1+\frac{6}{n}}+1}$. (1 pt)

3. Deduce the limit of (v_n). (1 pt)

4. Define $w_n = 2n^2 - 6n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this statement true? Justify. (1 pt)

Exercise 2 · True or false: reasoning about limits — 5 points

For each of the following statements, indicate whether it is true or false and justify the answer. An unjustified answer earns no marks. Limits of sequences are taken as n tends to $+\infty$.

1. If $u_n \to 5$ and $v_n \to -6$, then $\frac{u_n}{v_n} \to -\frac{5}{6}$. (1 pt)

2. The sequence $u_n = 5 + (-1)^n$ converges to 5. (1 pt)

3. If $u_n \to 0$ and $v_n \to +\infty$, then $u_n v_n \to 0$. (1 pt)

4. The sequence $w_n = \frac{\sqrt{n^2+n}}{n}$ for $n \geq 1$ converges to 1. (1 pt)

5. For $u_n = \frac{5n+6}{n}$, we have $|u_n - 5| < 0.01$ as soon as $n > 600$. (1 pt)

DS2 · Specimen 2 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+7\vec{\jmath}+11\vec{k}, \quad \vec{z}=4\vec{\imath}+7\vec{\jmath}+12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (1 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (1.5 pt)

4. Express the vector $\vec{u}-4\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (1 pt)

Exercise 4 · Geometry of a parallelepiped — 5 points

Consider a parallelepiped ABCDEFGH non-degenerate. The vertices E, F, G and H are the images respectively of A, B, C and D under the translation by vector $\overrightarrow{AE}$. Set $\vec{\imath}=\overrightarrow{AB}$, $\vec{\jmath}=\overrightarrow{AD}$ and $\vec{k}=\overrightarrow{AE}$. The points I, J and K are defined by

$$\overrightarrow{AI}=\frac{1}{8}\vec{\imath}, \qquad \overrightarrow{AJ}=\frac{1}{8}\vec{\jmath}, \qquad \overrightarrow{AK}=\frac{1}{8}\vec{k}.$$

1. Express $\overrightarrow{AG}$ and $\overrightarrow{BH}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

2. Prove that the lines (IJ) and (BD) are parallel. (1 pt)

3. Study the relative position of the lines (AB) and (CG) : are they parallel, intersecting or not coplanar? Justify. (1 pt)

4. Let O the parallelepiped's centre. Does point O lie in the plane (ABG) ? Justify. (1 pt)

5. Prove that the points I, J and K are not collinear. (1 pt)

Common test · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 2$ and, for every natural number n, $u_{n+1} = \frac{1}{4}u_n + 6$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Prove by induction that, for every $n \in \mathbb{N}$, $u_n = 8 - 6\left(\frac{1}{4}\right)^n$. (2 pt)

3. Deduce that, for every natural number n, $2 \leq u_n < 8$. (1 pt)

4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. A student claims: "There exists an integer n such that $u_n = 8$." Is this statement true? Justify. (0.5 pt)

Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{3n^2+5n-1}{4n^2+2}$ & $v_n = \sqrt{n^2+5n} - n$.

1. Determine the limit of (u_n) showing how its expression is transformed. (1 pt)

2. Show that, for every integer $n \geq 1$, $v_n = \frac{5}{\sqrt{1+\frac{5}{n}}+1}$. (1 pt)

3. Deduce the limit of (v_n). (1 pt)

4. Define $w_n = 3n^2 - 5n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this statement true? Justify. (1 pt)

Common test · Specimen 1 · continued

Exercise 3 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{2x+6}{x-2}$. Let $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the domain of f. Find two reals A and B such that, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (1 pt)
3. Determine the limits of f at 2^- and at 2^+. (1 pt)
4. Give the equations of the asymptotes to $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. (1 pt)
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^2 e^{-2x}$ with justification. (1 pt)

Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 4 symbols chosen from an alphabet that contains 15 distinct symbols. The rules concerning repetitions are specified in each question, independently of the other questions.

1. How many codes can be formed if repetitions are allowed? (1 pt)
2. How many codes can be formed if each symbol is to be used at most once? (1 pt)
3. One symbol of the alphabet is fixed in advance. With repetition allowed, how many codes contain this symbol at least once? (1 pt)
4. The first symbol of the code is fixed and repetitions are not allowed. How many codes can be formed? (1 pt)
5. Two distinct symbols of the alphabet are forbidden. The other symbols may be repeated. How many codes can be formed? (1 pt)

Common test · Specimen 2

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A discrete evolution — 5 points

Consider the sequence (u_n) defined by $u_0 = 3$ and, for every natural number n, $u_{n+1} = \frac{1}{3}u_n + \frac{20}{3}$.

1. Calculate u_1 and u_2. (0.5 pt)

2. Prove by induction that, for every $n \in \mathbb{N}$, $u_n = 10 - 7\left(\frac{1}{3}\right)^n$. (2 pt)

3. Deduce that, for every natural number n, $3 \leq u_n < 10$. (1 pt)

4. Study the sign of $u_{n+1} - u_n$ and deduce the direction of variation of (u_n). (1 pt)

5. A student claims: "There exists an integer n such that $u_n = 10$." Is this statement true? Justify. (0.5 pt)

Exercise 2 · Limits calculated and justified — 5 points

For every integer $n \geq 1$, we define $u_n = \frac{2n^2+7n-1}{3n^2+3}$ & $v_n = \sqrt{n^2+7n} - n$.

1. Determine the limit of (u_n) showing how its expression is transformed. (1 pt)

2. Show that, for every integer $n \geq 1$, $v_n = \frac{7}{\sqrt{1+\frac{7}{n}}+1}$. (1 pt)

3. Deduce the limit of (v_n). (1 pt)

4. Define $w_n = 2n^2 - 7n$ for $n \geq 1$. Determine the limit of (w_n). (1 pt)

5. A student claims that the difference of two sequences tending to $+\infty$ always tends to 0. Is this statement true? Justify. (1 pt)

Common test · Specimen 2 · continued

Exercise 3 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{5x+7}{x-3}$. Let $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the domain of f. Find two reals A and B such that, for all $x \neq 3$, $f(x) = A + \frac{B}{x-3}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (1 pt)
3. Determine the limits of f at 3^- and at 3^+. (1 pt)
4. Give the equations of the asymptotes to $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. (1 pt)
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^5 e^{-3x}$ with justification. (1 pt)

Exercise 4 · Codes and selections — 5 points

A code is an ordered sequence of 5 symbols chosen from an alphabet that contains 16 distinct symbols. The rules concerning repetitions are specified in each question, independently of the other questions.

1. How many codes can be formed if repetitions are allowed? (1 pt)
2. How many codes can be formed if each symbol is to be used at most once? (1 pt)
3. One symbol of the alphabet is fixed in advance. With repetition allowed, how many codes contain this symbol at least once? (1 pt)
4. The first symbol of the code is fixed and repetitions are not allowed. How many codes can be formed? (1 pt)
5. Two distinct symbols of the alphabet are forbidden. The other symbols may be repeated. How many codes can be formed? (1 pt)

DS3 · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{5x+4}{x-1}$. Let $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the domain of f. Find two reals A and B such that, for all $x \neq 1$, $f(x) = A + \frac{B}{x-1}$. (1 pt)

2. Determine the limits of f at $-\infty$ and at $+\infty$. (1 pt)

3. Determine the limits of f at 1^- and at 1^+. (1 pt)

4. Give the equations of the asymptotes to $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. (1 pt)

5. Independently of the previous questions, determine $\lim_{x \to +\infty} x^5 e^{-1x}$ with justification. (1 pt)

Exercise 2 · Variations, convexity and equation — 5 points

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^3 + 4x - 5$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$, then study the direction of variation of f on $\mathbb{R}$. (1 pt)

2. Determine the limits of f at $-\infty$ and at $+\infty$. (0.5 pt)

3. Prove that the equation $f(x) = 0$ has a unique real solution α. Give its exact value if it is an integer; otherwise, bound it between two consecutive integers. (1.5 pt)

4. Study the convexity of f, then find the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0, then study the position of the curve relative to this tangent. (1 pt)

DS3 · Specimen 1 · continued

Exercise 3 · An exponential curve — 5 points

Consider the function f defined on $[0;+\infty[$ by $f(x) = (x+3)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Prove that the equation $f(x) = \frac{3}{2}$ has a unique solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Forming a group — 5 points

A team consists of 6 teachers and 10 students, all distinct. A group of four people is chosen; the order of choice does not matter.

1. How many groups of four people can be formed? (1 pt)
2. How many of these groups include exactly two teachers? (1 pt)
3. How many of these groups include at least one teacher and at least one student? (1 pt)
4. We choose a group containing exactly two teachers, then appoint from among its members a teacher spokesperson and a student rapporteur. Count the choices of the group together with these two assignments. (1 pt)
5. Explain why the number of ordered lists of four distinct people, divided by $4!$, gives the number of groups obtained in question 1. (1 pt)

DS3 · Specimen 2

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Limits and asymptotes — 5 points

Consider the function f defined by $f(x) = \frac{4x+6}{x-2}$. Let $\mathcal{C}_f$ its representative curve in a coordinate system in the plane.

1. Determine the domain of f. Find two reals A and B such that, for all $x \neq 2$, $f(x) = A + \frac{B}{x-2}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (1 pt)
3. Determine the limits of f at 2^- and at 2^+. (1 pt)
4. Give the equations of the asymptotes to $\mathcal{C}_f$. State the position of the curve relative to its horizontal asymptote. (1 pt)
5. Independently of the previous questions, determine $\lim\limits_{x \to +\infty} x^4 e^{-2x}$ with justification. (1 pt)

Exercise 2 · Variations, convexity and equation — 5 points

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^3 + 3x - 7$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$, then study the direction of variation of f on $\mathbb{R}$. (1 pt)
2. Determine the limits of f at $-\infty$ and at $+\infty$. (0.5 pt)
3. Prove that the equation $f(x) = 0$ has a unique real solution α. Give its exact value if it is an integer; otherwise, bound it between two consecutive integers. (1.5 pt)
4. Study the convexity of f, then find the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0, then study the position of the curve relative to this tangent. (1 pt)

DS3 · Specimen 2 · continued

Exercise 3 · An exponential curve — 5 points

Consider the function f defined on $[0;+\infty[$ by $f(x) = (x+2)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)
2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)
3. Calculate $f''(x)$, then study the convexity of f. (1 pt)
4. Prove that the equation $f(x) = 1$ has a unique solution α in $[0;+\infty[$. (1.5 pt)
5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (1 pt)

Exercise 4 · Forming a group — 5 points

A team consists of 9 teachers and 11 students, all distinct. A group of four people is chosen; the order of choice does not matter.

1. How many groups of four people can be formed? (1 pt)
2. How many of these groups include exactly two teachers? (1 pt)
3. How many of these groups include at least one teacher and at least one student? (1 pt)
4. We choose a group containing exactly two teachers, then appoint from among its members a teacher spokesperson and a student rapporteur. Count the choices of the group together with these two assignments. (1 pt)
5. Explain why the number of ordered lists of four distinct people, divided by $4!$, gives the number of groups obtained in question 1. (1 pt)

Mock exam 1 · Zero 1

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{42 + v_n}$.

1. Calculate v_1 and v_2. Give exact values. (0.25 pt)

2. Prove by induction that all terms are defined and that, for every $n \in \mathbb{N}$, $0 \leq v_n \leq 7$. (0.75 pt)

3. Prove by induction that, for every $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (0.75 pt)

4. Prove that, for every $n \in \mathbb{N}$, $v_n < 7$. (0.5 pt)

5. Justify that (v_n) converges, then determine its limit. (0.75 pt)

6. Prove that, for every $n \in \mathbb{N}$, $0 \leq 7 - v_{n+1} \leq \frac{7 - v_n}{7}$. (0.5 pt)

7. Deduce by induction an upper bound for $7 - v_n$ as a function of n. Then determine, by comparison of integer powers, an index from which this upper bound guarantees $7 - v_n \leq 10^{-3}$. (1 pt)

8. Write a Python function with no argument that returns the first integer n such that $7 - v_n \leq 10^{-3}$. Run the function and compare the index obtained with the sufficient index from the previous question. Explain any difference. (0.5 pt)

Exercise 2 · An exponential curve — 5 points

Consider the function f defined on $[0; +\infty[$ by $f(x) = (x + 5)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Prove that the equation $f(x) = \frac{5}{2}$ has a unique solution α in $[0; +\infty[$. (1 pt)

5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0; 2]$ and returns an interval containing α of width at most 10^{-3}. (0.5 pt)

7. Determine the minimum number of bisection steps guaranteeing this width. (0.5 pt)

8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the x-axis." Refute this statement by studying the sign of $f(x)$. (1 pt)

Mock exam 1 · Zero 1 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=4\vec{\imath}+6\vec{\jmath}+10\vec{k}, \quad \vec{z}=4\vec{\imath}+6\vec{\jmath}+11\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (0.5 pt)

4. Express the vector $6\vec{u}-4\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances AB, AC and BC. (0.5 pt)

7. Let I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are collinear. (0.5 pt)

8. Define E by $\overrightarrow{AE}=\vec{z}$. Is the line (DE) contained in the plane (ABC) ? Justify. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 19 times. The trials are independent and the probability of success is $p=\frac{3}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N\geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j\in\{0,\dots,19\}$, prove the formula for $P(X=j)$ by counting the possible positions of successes. (0.5 pt)

7. A series of 19 trials is accepted if its number of successes is between 1 and 3, endpoints included. Calculate the probability that a series is accepted. (0.5 pt)

8. Write a Python function with no argument that returns the smallest integer j such that $P(X\leq j)\geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Mock exam 1 · Zero 2

SPECIMEN PAPER · PRACTICE

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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Induction and bounds — 5 points

Consider the sequence (v_n) defined by $v_0 = 0$ and, for every natural number n, $v_{n+1} = \sqrt{110 + v_n}$.

1. Calculate v_1 and v_2. Give exact values. (0.25 pt)

2. Prove by induction that all terms are defined and that, for every $n \in \mathbb{N}$, $0 \leq v_n \leq 11$. (0.75 pt)

3. Prove by induction that, for every $n \in \mathbb{N}$, $v_{n+1} \geq v_n$. (0.75 pt)

4. Prove that, for every $n \in \mathbb{N}$, $v_n < 11$. (0.5 pt)

5. Justify that (v_n) converges, then determine its limit. (0.75 pt)

6. Prove that, for every $n \in \mathbb{N}$, $0 \leq 11 - v_{n+1} \leq \frac{11 - v_n}{11}$. (0.5 pt)

7. Deduce by induction an upper bound for $11 - v_n$ as a function of n. Then determine, by comparison of integer powers, an index from which this upper bound guarantees $11 - v_n \leq 10^{-3}$. (1 pt)

8. Write a Python function with no argument that returns the first integer n such that $11 - v_n \leq 10^{-3}$. Run the function and compare the index obtained with the sufficient index from the previous question. Explain any difference. (0.5 pt)

Exercise 2 · An exponential curve — 5 points

Consider the function f defined on $[0; +\infty[$ by $f(x) = (x + 4)e^{-x}$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f(0)$ and determine the limit of f at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. Calculate $f''(x)$, then study the convexity of f. (0.5 pt)

4. Prove that the equation $f(x) = 2$ has a unique solution α in $[0; +\infty[$. (1 pt)

5. Determine an equation of the tangent to $\mathcal{C}_f$ at the point with x-coordinate 0. Using convexity, give an inequality between $f(x)$ and the affine expression of this tangent, for all $x \geq 0$. (0.5 pt)

6. Show that $0 < \alpha < 2$. Write a Python function that applies bisection on $[0; 2]$ and returns an interval containing α of width at most 10^{-3}. (0.5 pt)

7. Determine the minimum number of bisection steps guaranteeing this width. (0.5 pt)

8. A student claims: "Since $f(x)$ tends to 0, the curve crosses the x-axis." Refute this statement by studying the sign of $f(x)$. (1 pt)

Mock exam 1 · Zero 2 · continued

Exercise 3 · Vector calculation in a basis — 5 points

In a basis $(\vec{\imath},\vec{\jmath},\vec{k})$ of space, we define the vectors

$$\vec{u}=\vec{\imath}+\vec{k}, \qquad \vec{v}=\vec{\jmath}+\vec{k},$$
$$\vec{w}=3\vec{\imath}+8\vec{\jmath}+11\vec{k}, \quad \vec{z}=3\vec{\imath}+8\vec{\jmath}+12\vec{k}.$$

1. Justify that $\vec{u}$ and $\vec{v}$ are not collinear. (0.5 pt)

2. Write $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$. (0.5 pt)

3. The vectors $\vec{u}$, $\vec{v}$ and $\vec{z}$ coplanar? Justify. (0.5 pt)

4. Express the vector $8\vec{u}-3\vec{v}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$. (1 pt)

5. We choose a point A, then we define B, C and D by $\overrightarrow{AB}=\vec{u}$, $\overrightarrow{AC}=\vec{v}$ and $\overrightarrow{AD}=\vec{w}$. Prove that A, B, C and D are coplanar. (0.5 pt)

6. For questions 6 to 8, assume the basis is orthonormal. Calculate the distances AB, AC and BC. (0.5 pt)

7. Let I the midpoint of $[BC]$. Express $\overrightarrow{AI}$ in the basis $(\vec{\imath},\vec{\jmath},\vec{k})$, then determine whether the points A, I and D are collinear. (0.5 pt)

8. Define E by $\overrightarrow{AE}=\vec{z}$. Is the line (DE) contained in the plane (ABC) ? Justify. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 16 times. The trials are independent dent and the probability of success is $p=\frac{1}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N\geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j\in\{0,\dots,16\}$, prove the formula for $P(X=j)$ by counting the possible positions of successes. (0.5 pt)

7. A series of 16 trials is accepted if its number of successes is between 1 and 3, endpoints included. Calculate the probability that a series is accepted. (0.5 pt)

8. Write a Python function with no argument that returns the smallest integer j such that $P(X\leq j)\geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

DS4 · Specimen 1

SPECIMEN PAPER · PRACTICE

Duration: 2 h · Marked out of 20 · Four independent exercises.
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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{3+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (1 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)

3. State the maximum value of f, then solve $f(x)=0$. (1 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x)=\frac{1}{2}e^2$ with justification. $]0;+\infty[$. (1 pt)

Exercise 2 · Antiderivatives and an imposed condition — 5 points

The first four questions are independent. For each, give the expression of the required antiderivatives and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 2x^2-6x+2$. (1 pt)

2. Determine the antiderivative F of $x\mapsto e^{2x}$ on $\mathbb{R}$ such that $F(0)=6$. (1 pt)

3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+6}$. (1 pt)

4. Determine an antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+2)^2$. (1 pt)

5. Let F and G two antiderivatives of the same function over an interval I. Prove that there exists a real constant C such that $F(x)-G(x)=C$ for all $x\in I$. (1 pt)

DS4 · Specimen 1 · continued

Exercise 3 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{5}y + \frac{8}{5}$ on $[0; +\infty[$. In questions 2 to 5, we call y the solution that satisfies $y(0) = 13$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution satisfying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ as t tends to $+\infty$. (1 pt)

4. Study the direction of variation of y, then find the smallest real number $t \geq 0$ such that $y(t) \leq 9$. (1 pt)

5. Euler's method is applied with a step $h > 0$ and let Y_k call the approx. of $y(kh)$. Give the value of Y_0 and express Y_{k+1} as a function of Y_k and of h. (1 pt)

Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, we consider $A(11;0;0)$, $B(0;11;0)$ and $C(0;0;11)$. Let $\vec{n} = (1;1;1)$. Lengths, areas and volumes are expressed in units of length, of area and volume.

1. Prove that $\vec{n}$ is normal to the plane (ABC). (1 pt)

2. Let H the orthogonal projection of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Prove that OH is the distance from O to the plane (ABC) : for every $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the area of the triangle ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

DS4 · Specimen 2

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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{2+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (1 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (1 pt)

3. State the maximum value of f, then solve $f(x)=0$. (1 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x)=\frac{1}{2}e^1$ with justification. $]0;+\infty[$. (1 pt)

Exercise 2 · Antiderivatives and an imposed condition — 5 points

The first four questions are independent. For each, give the expression of the required antiderivatives and check the result by differentiation.

1. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto 5x^2-7x+3$. (1 pt)

2. Determine the antiderivative F of $x\mapsto e^{5x}$ on $\mathbb{R}$ such that $F(0)=7$. (1 pt)

3. Determine all antiderivatives on $\mathbb{R}$ of $x\mapsto\frac{2x}{x^2+7}$. (1 pt)

4. Determine an antiderivative on $\mathbb{R}$ of $x\mapsto 2x(x^2+3)^5$. (1 pt)

5. Let F and G two antiderivatives of the same function over an interval I. Prove that there exists a real constant C such that $F(x)-G(x)=C$ for all $x\in I$. (1 pt)

DS4 · Specimen 2 · continued

Exercise 3 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{4}y + \frac{5}{2}$ on $[0; +\infty[$. In questions 2 to 5, we call y the solution that satisfies $y(0) = 16$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution satisfying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ as t tends to $+\infty$. (1 pt)

4. Study the direction of variation of y, then find the smallest real number $t \geq 0$ such that $y(t) \leq 11$. (1 pt)

5. Euler's method is applied with a step $h > 0$ and let Y_k call the approx. of $y(kh)$. Give the value of Y_0 and express Y_{k+1} as a function of Y_k and of h. (1 pt)

Exercise 4 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, we consider $A(7;0;0)$, $B(0;7;0)$ and $C(0;0;7)$. Let $\vec{n} = (1;1;1)$. Lengths, areas and volumes are expressed in units of length, of area and volume.

1. Prove that $\vec{n}$ is normal to the plane (ABC). (1 pt)

2. Let H the orthogonal projection of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Prove that OH is the distance from O to the plane (ABC) : for every $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the area of the triangle ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

DS5 · Specimen 1

SPECIMEN PAPER · PRACTICE

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Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, set $I_n = \int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{4x}$.

1. Calculate I_0. (1 pt)

2. Using integration by parts, calculate I_1. (1 pt)

3. Prove, by comparing the integrands, that (I_n) is positive and decreasing. (1 pt)

4. Show that $I_n \leq \frac{e^4}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). (1 pt)

5. Using integration by parts, express I_{n+1} as a function of I_n and of n. (1 pt)

Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{3}y + \frac{8}{3}$ on $[0;+\infty[$. In questions 2 to 5, we call y the solution that satisfies $y(0) = 12$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution satisfying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ as t tends to $+\infty$. (1 pt)

4. Study the direction of variation of y, then find the smallest real number $t \geq 0$ such that $y(t) \leq 9$. (1 pt)

5. Euler's method is applied with a step $h > 0$ and let Y_k call the approx. of $y(kh)$. Give the value of Y_0 and express Y_{k+1} as a function of Y_k and of h. (1 pt)

DS5 · Specimen 1 · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, we consider $A(9;0;0)$, $B(0;9;0)$ and $C(0;0;9)$. Let $\vec{n} = (1;1;1)$. Lengths, areas and volumes are expressed in units of length, of area and volume.

1. Prove that $\vec{n}$ is normal to the plane (ABC). (1 pt)

2. Let H the orthogonal projection of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Prove that OH is the distance from O to the plane (ABC) : for every $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the area of the triangle ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · Sums of random variables — 5 points

Let $n = 17$. Consider n independent random variables $X_1, \ldots, X_{17}$, which all follow the Bernoulli distribution with parameter $p = \frac{1}{5}$. Set $S = \sum_{i=1}^{17} X_i$ and $G = 9S - 17$.

1. Justify the distribution of S and calculate $E(S)$. (1 pt)

2. From the variance of a Bernoulli variable and independence, calculate $V(S)$. (1 pt)

3. Calculate $E(G)$ and $V(G)$. (1 pt)

4. Set $M = \frac{S}{17}$. Calculate $E(M)$ and $V(M)$. (1 pt)

5. For this question only, the 17 variables are replaced by 17 variables all equal to X_1. Then calculate the variance of their mean and compare it with that of question 4. Explain why the results differ. (1 pt)

DS5 · Specimen 2

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Exercise 1 · Integral and sequence — 5 points

For every natural number n, set $I_n = \int_0^1 x^n e^{3x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{3x}$.

1. Calculate I_0. (1 pt)

2. Using integration by parts, calculate I_1. (1 pt)

3. Prove, by comparing the integrands, that (I_n) is positive and decreasing. (1 pt)

4. Show that $I_n \leq \frac{e^3}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). (1 pt)

5. Using integration by parts, express I_{n+1} as a function of I_n and of n. (1 pt)

Exercise 2 · Differential equation and evolution — 5 points

Consider the differential equation $(E) : y' = -\frac{1}{2}y + 5$ on $[0; +\infty[$. In questions 2 to 5, we call y the solution that satisfies $y(0) = 15$.

1. Determine the constant solution of (E). (1 pt)

2. Determine the expression of the solution satisfying the given initial condition. (1 pt)

3. Determine the limit of $y(t)$ as t tends to $+\infty$. (1 pt)

4. Study the direction of variation of y, then find the smallest real number $t \geq 0$ such that $y(t) \leq 11$. (1 pt)

5. Euler's method is applied with a step $h > 0$ and let Y_k call the approx. of $y(kh)$. Give the value of Y_0 and express Y_{k+1} as a function of Y_k and of h. (1 pt)

DS5 · Specimen 2 · continued

Exercise 3 · Projection and volume — 5 points

In an orthonormal coordinate system with origin O, we consider $A(13;0;0)$, $B(0;13;0)$ and $C(0;0;13)$. Let $\vec{n} = (1;1;1)$. Lengths, areas and volumes are expressed in units of length, of area and volume.

1. Prove that $\vec{n}$ is normal to the plane (ABC). (1 pt)

2. Let H the orthogonal projection of O on (ABC). Determine the coordinates of H using $\overrightarrow{AH} \cdot \vec{n} = 0$. (1 pt)

3. Prove that OH is the distance from O to the plane (ABC) : for every $P \in (ABC)$, establish $OP^2 \geq OH^2$. (1 pt)

4. Calculate the area of the triangle ABC. (1 pt)

5. Calculate the volume of the tetrahedron $OABC$ in two ways, choosing two different bases. *(1 pt)*

Exercise 4 · Sums of random variables — 5 points

Let $n = 14$. Consider n independent random variables $X_1, \ldots, X_{14}$, which all follow the Bernoulli distribution with parameter $p = \frac{1}{4}$. Set $S = \sum_{i=1}^{14} X_i$ and $G = 6S - 14$.

1. Justify the distribution of S and calculate $E(S)$. (1 pt)

2. From the variance of a Bernoulli variable and independence, calculate $V(S)$. (1 pt)

3. Calculate $E(G)$ and $V(G)$. (1 pt)

4. Set $M = \frac{S}{14}$. Calculate $E(M)$ and $V(M)$. (1 pt)

5. For this question only, the 14 variables are replaced by 14 variables all equal to X_1. Then calculate the variance of their mean and compare it with that of question 4. Explain why the results differ. (1 pt)

Mock exam 2 · Zero 1

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, set $I_n = \int_0^1 x^n e^{5x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{5x}$.

1. Calculate I_0. (0.5 pt)

2. Using integration by parts, calculate I_1. (0.5 pt)

3. Prove, by comparing the integrands, that (I_n) is positive and decreasing. (0.5 pt)

4. Show that $I_n \leq \frac{e^5}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). (1 pt)

5. Using integration by parts, express I_{n+1} as a function of I_n and of n. (0.5 pt)

6. Show that $I_n \geq \frac{1}{n+1}$ for all $n \in \mathbb{N}$. (0.5 pt)

7. We divide $[0;1]$ in N intervals of the same length, with $N \geq 1$. Write the sums L_N and U_N of areas of the rectangles built with the values of e^{5x} at the left and right ends. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. (0.5 pt)

8. Determine the smallest integer N guaranteeing that this bracketing has a width at most equal to 10^{-2}. Write a Python function that returns N, L_N and U_N. (1 pt)

Exercise 2 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x) = \frac{4+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. State the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x) = \frac{1}{2}e^3$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the mean value of f on $[1;e]$. (0.5 pt)

8. Independently of the previous questions, solve $\ln(x-3) + \ln(x+3) = \ln(27)$, after specifying the set on which this equation is defined. (1 pt)

Mock exam 2 · Zero 1 · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 10$ and the point $M(3;0;0)$. The line d passes through M and has direction vector $\vec{n} = (1;2;2)$.

1. Give a parametric representation of d. (0.5 pt)

2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal projection of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ through M. (0.5 pt)

7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 20 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 20\}$, prove the formula for $P(X = j)$ by counting the possible positions of successes. (0.5 pt)

7. A series of 20 trials is accepted if its number of successes is between 1 and 3, endpoints included. Calculate the probability that a series is accepted. (0.5 pt)

8. Write a Python function with no argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Mock exam 2 · Zero 2

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · Integral and sequence — 5 points

For every natural number n, set $I_n = \int_0^1 x^n e^{4x}\,\mathrm{d}x$. For $n = 0$, the integrand is the function $x \mapsto e^{4x}$.

1. Calculate I_0. (0.5 pt)

2. Using integration by parts, calculate I_1. (0.5 pt)

3. Prove, by comparing the integrands, that (I_n) is positive and decreasing. (0.5 pt)

4. Show that $I_n \leq \frac{e^4}{n+1}$ for all $n \in \mathbb{N}$. Deduce the limit of (I_n). (1 pt)

5. Using integration by parts, express I_{n+1} as a function of I_n and of n. (0.5 pt)

6. Show that $I_n \geq \frac{1}{n+1}$ for all $n \in \mathbb{N}$. (0.5 pt)

7. We divide $[0;1]$ in N intervals of the same length, with $N \geq 1$. Write the sums L_N and U_N of areas of the rectangles built with the values of e^{4x} at the left and right ends. Justify $L_N \leq I_0 \leq U_N$, then calculate $U_N - L_N$. (0.5 pt)

8. Determine the smallest integer N guaranteeing that this bracketing has a width at most equal to 10^{-2}. Write a Python function that returns N, L_N and U_N. (1 pt)

Exercise 2 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x) = \frac{3+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. State the maximum value of f, then solve $f(x) = 0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x) = \frac{1}{2}e^2$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the mean value of f on $[1;e]$. (0.5 pt)

8. Independently of the previous questions, solve $\ln(x-1) + \ln(x+1) = \ln(63)$, after specifying the set on which this equation is defined. (1 pt)

Mock exam 2 · Zero 2 · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 6$ and the point $M(1;0;0)$. The line d passes through M and has direction vector $\vec{n} = (1;2;2)$.

1. Give a parametric representation of d. (0.5 pt)

2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal projection of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ through M. (0.5 pt)

7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 17 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j \in \{0, \dots, 17\}$, prove the formula for $P(X = j)$ by counting the possible positions of successes. (0.5 pt)

7. A series of 17 trials is accepted if its number of successes is between 1 and 3, endpoints included. Calculate the probability that a series is accepted. (0.5 pt)

8. Write a Python function with no argument that returns the smallest integer j such that $P(X \leq j) \geq 0{,}95$. Run the function and check the minimality of the result. (1 pt)

Final mock exam · Zero 1

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{2+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. State the maximum value of f, then solve $f(x)=0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x)=\frac{1}{2}e^1$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the mean value of f on $[1;e]$. (0.5 pt)

8. Independently of the previous questions, solve $\ln(x-2)+\ln(x+2)=\ln(32)$, after specifying the set on which this equation is defined. (1 pt)

Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel turning at constant speed in a vertical plane. Its height above the ground is modelled by $h(t)=9-3\cos\left(\frac{\pi t}{5}\right)$, for t expressed in seconds and $h(t)$ in metres. We study one complete revolution, for $t\in[0;10]$.

1. Give a period of the model extended to $\mathbb{R}$, then the minimum and maximum heights on a turn. (0.5 pt)

2. Calculate $h'(t)$ and draw up the table of variations of h on $[0;10]$. (0.5 pt)

3. Determine all the times $t\in[0;10]$ such that $h(t)=\frac{21}{2}$. (0.5 pt)

4. Calculate the mean value of h on $[0;10]$. (1 pt)

5. Study the convexity of h on the interval studied and determine the coordinates of the points of inflection of its curve. (0.5 pt)

6. Determine the total time during which the height is at least the level of question 3. (0.5 pt)

7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, as well as the times at which they are reached. (0.5 pt)

8. Calculate the vertical displacement $h(10)-h(0)$, then the sum of the height climbed on the way up and the height descended on the way down. Explain why these two results are different. (1 pt)

Final mock exam · Zero 1 · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 11$ and point $M(2;0;0)$. The line d passes through M and has direction vector $\vec{n} = (1;2;2)$.

1. Give a parametric representation of d. (0.5 pt)

2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal projection of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ through M. (0.5 pt)

7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 20 times. The trials are independent and the probability of success is $p = \frac{1}{5}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j \in \{0, \ldots, 20\}$, prove the formula for $P(X = j)$ by counting the possible positions of successes. (0.5 pt)

7. For N independent trials with the same parameter p, let F_N be the frequency of successes. Prove the expressions for $E(F_N)$ and $V(F_N)$ as a function of N and p. (0.5 pt)

8. Using Bienaymé-Chebyshev, determine a sample size N sufficient to ensure $P(|F_N - p| < 0{,}02) \geq 0{,}95$. Is the size obtained necessarily the smallest possible for this probability ? Justify. (1 pt)

Final mock exam · Zero 2

SPECIMEN PAPER · PRACTICE

Duration: 4 h · Marked out of 20 · Four independent exercises.
Calculator allowed. Every answer must be justified; incomplete work will be taken into account. Give exact values; round probabilities to 10^{-6} and the other approximations to 10^{-4}, unless otherwise specified.

Name: ..... **Class:** ..... **Date:** .....

Exercise 1 · A logarithmic function — 5 points

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{5+\ln x}{x}$. Let $\mathcal{C}_f$ its representative curve.

1. Determine the limits of f at 0^+ and at $+\infty$. (0.5 pt)

2. Calculate $f'(x)$, then draw up the table of variations of f. (0.5 pt)

3. State the maximum value of f, then solve $f(x)=0$. (0.5 pt)

4. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$. (1 pt)

5. Prove that the equation $f(x)=\frac{1}{2}e^4$ has exactly two solutions in $]0;+\infty[$. (0.5 pt)

6. Determine an antiderivative of f on $]0;+\infty[$. (0.5 pt)

7. Calculate the mean value of f on $[1;e]$. (0.5 pt)

8. Independently of the previous questions, solve $\ln(x-3)+\ln(x+3)=\ln(40)$, after specifying the set on which this equation is defined. (1 pt)

Exercise 2 · Periodic phenomenon — 5 points

A point moves on a wheel turning at constant speed in a vertical plane. Its height above the ground is modelled by $h(t)=12-4\cos\left(\frac{\pi t}{4}\right)$, for t expressed in seconds and $h(t)$ in metres. We study one complete revolution, for $t\in[0;8]$.

1. Give a period of the model extended to $\mathbb{R}$, then the minimum and maximum heights on a turn. (0.5 pt)

2. Calculate $h'(t)$ and draw up the table of variations of h on $[0;8]$. (0.5 pt)

3. Determine all the times $t\in[0;8]$ such that $h(t)=14$. (0.5 pt)

4. Calculate the mean value of h on $[0;8]$. (1 pt)

5. Study the convexity of h on the interval studied and determine the coordinates of the points of inflection of its curve. (0.5 pt)

6. Determine the total time during which the height is at least the level of question 3. (0.5 pt)

7. The vertical speed is given by $h'(t)$. Determine its maximum and minimum values, in metres per second, as well as the times at which they are reached. (0.5 pt)

8. Calculate the vertical displacement $h(8)-h(0)$, then the sum of the height climbed on the way up and the height descended on the way down. Explain why these two results are different. (1 pt)

Final mock exam · Zero 2 · continued

Exercise 3 · Line, plane and projection — 5 points

In an orthonormal coordinate system of space, consider the plane $\mathcal{P}$ with equation $x + 2y + 2z = 7$ and point $M(3;0;0)$. The line d passes through M and has direction vector $\vec{n} = (1;2;2)$.

1. Give a parametric representation of d. (0.5 pt)

2. Prove that d meets $\mathcal{P}$ at a point H, then calculate the coordinates of H. (0.5 pt)

3. Justify that H is the orthogonal projection of M on $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$. (0.5 pt)

4. Consider the plane $\mathcal{Q}$ with equation $y = z$. Prove that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it. (1 pt)

5. Verify that $H \in \mathcal{Q}$. Deduce the distance from M to the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

6. Determine a Cartesian equation of the plane parallel to $\mathcal{P}$ through M. (0.5 pt)

7. Determine the angle between d and the line $\mathcal{P} \cap \mathcal{Q}$. (0.5 pt)

8. Define the point N by $\overrightarrow{MN} = \vec{n}$. Calculate the distances from N to the plane $\mathcal{P}$ and to the line $\mathcal{P} \cap \mathcal{Q}$, then compare them. (1 pt)

Exercise 4 · Repetitions and estimation — 5 points

A trial with two outcomes, called success and failure, is repeated 17 times. The trials are independent and the probability of success is $p = \frac{3}{10}$ at each trial. Let X be the number of successes obtained.

1. Justify that X follows a binomial distribution and state its parameters. (0.5 pt)

2. Calculate the probability of exactly two successes. (0.5 pt)

3. Using complementary events, express and then calculate the probability of obtaining at least two successes. (0.5 pt)

4. Calculate the expectation and variance of X. (1 pt)

5. We now repeat the same trial $N \geq 1$ times, still independently. Determine the smallest integer N for which the probability of getting at least one success is greater than or equal to 0.95. A search using a table or a loop is allowed. (0.5 pt)

6. For an integer $j \in \{0, \dots, 17\}$, prove the formula for $P(X = j)$ by counting the possible positions of successes. (0.5 pt)

7. For N independent trials with the same parameter p, let F_N be the frequency of successes. Prove the expressions for $E(F_N)$ and $V(F_N)$ as a function of N and p. (0.5 pt)

8. Using Bienaymé-Chebyshev, determine a sample size N sufficient to ensure $P(|F_N - p| < 0{,}02) \geq 0{,}95$. Is the size obtained necessarily the smallest possible for this probability ? Justify. (1 pt)